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EXTENDED ABSTRACT

AI-boosted rare event sampling to characterize extreme weather

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Introduction & Objectives

Assessing the frequency and intensity of extreme weather events, and understanding how climate change affects them, is crucial for developing effective adaptation and mitigation strategies (*1*–*5*). However, observational datasets are too short and physics-based global climate models (GCMs) are too computationally expensive to obtain robust statistics for the rarest, yet most impactful, extreme events (*6*). AI-based emulators have shown promise for predictions at weather and even climate timescales (*7*–*13*), but they struggle on extreme events with few, or no, examples in their training datasets (*14*–*17*). Rare event sampling (RES) algorithms, a set of importance sampling tools that focus computation on the rare event of interest, have previously demonstrated success for some extreme events (*18*–*23*), but their performance depends critically on a hard-to-identify "score function", which guides efficient sampling by a GCM.

Methods & Data

In this paper, we leverage advances in AI weather emulation to build an effective score function for RES, combining recent developments in both areas to construct a robust algorithm to characterize rare, extreme events (Fig. 1). As a demonstration of this new framework (AI+RES), we focus on simulating extreme heat waves and apply our approach to a GCM of intermediate complexity, PlaSim (*24*), which allows for extensive long-term control runs to compute baseline rare event statistics and rigorously evaluate the performance of our approach. Heat waves are a well-motivated application area because they are the deadliest extreme weather events and are expected to worsen under climate change (*25*).

Main Results

We show that AI+RES provides accurate rare event statistics at a numerical speed-up of up to a factor of $O(100)$, whereas standard RES fails to even yield examples of the rarest events of interest. This paper

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serves as a demonstration of a novel methodology that can overcome the primary difficulty of RES and unleash its power widely across any field for which an AI emulator can be constructed.

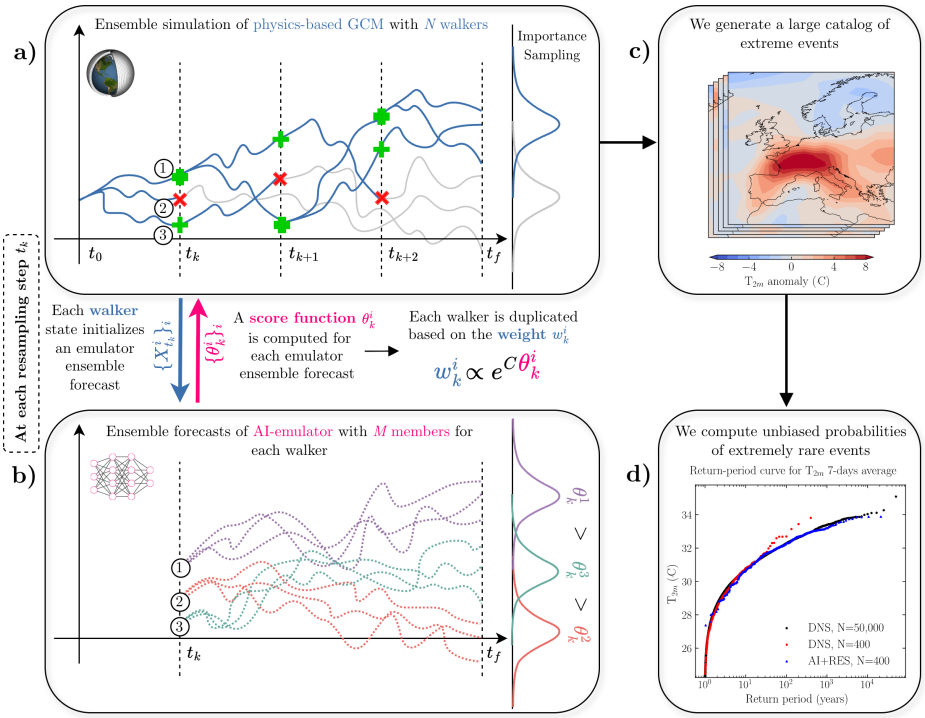

Figure 1. Schematic of the AI+RES framework. *We train an AI weather emulator on 100 years of PlaSim GCM data, then use it to guide a RES algorithm. (a) In our proposed method (AI+RES), we run a PlaSim ensemble simulation with N parallel simulations, called* walkers *and denoted* X_t^i*. (b) At each resampling time* t_k *in the algorithm (vertical dashed lines), we perform an ensemble forecast with the AI emulator for each PlaSim walker until the target time* t_f*, then duplicate the more promising ones based on these forecasts. This allows us to both generate a large catalog of extreme events from physics-based simulations (c) and to compute unbiased probabilities for these rare events (d) at a fraction of the cost of Direct Numerical Simulation (DNS). .*

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